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The spectral and temporal description of Javanese Gong Kempul

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In Central Java, the Gong Kempul is one of eminent gamelan instrument, an ensemble of predominantly struck instruments found across Central Java and has deep philosophica... **View more**

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Abstract:
In Central Java, the Gong Kempul is one of eminent gamelan instrument, an ensemble of predominantly struck instruments found across Central Java and has deep philosophical meaning for Javanese. However, there lack of studies concerning on this particular instrument as a bridging means between scientific description and human artistic perception. This study aims to investigate the spectral and temporal properties as well as particularly look into the typical wave-like sound of the Gong Kempul. Acoustic measurements were conducted and analyzed using ARTA. Both frequency and time domain analyses were explored to better understand the nature of the Gong Kempul. It was found that the partials at 93.8 Hz, 187.5 Hz, 281.3 Hz, and 375 Hz are the strict integer multiples of the fundamental frequency. It can also be inferred that the fundamental frequency at 93.8 Hz decays much more slowly than the other harmonic. In addition, the pitch of the Gong Kempul slide upward as the time goes on. The wave-like sound of the Gong Kempul is due to the beat phenomenon between two partials that occur closely.

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Contents

SECTION I.
Introduction

Gongs are instruments in Gamelan which perform what is often termed a colotomic function in the music to mark off established time intervals [1]– [2]. The smallest gongs which hang in a crossbar supported by two side posts are the Gong Kempul, as shown in Figure 1, with 56 cm in diameter. This instrument has a very deep and distinctly pitched roar mimicking the rolling waves of the sea. Sacrilegious aspect of Javanese Gamelan is very obvious, as in [3], and ones need a certain extent of

the scientific counterpart. There have been studies in the past that attempt to describe the spectrum

of gamelan instruments. Suprapto et al. [4] proposed to construct gamelan models. The main target of his research was creating Gamelan Frequency Modeling. He proposed two Frequency Balungan Models, the first model was using average value, and the other was using average value in the most dense area. Tjahyanto et al. [5] proposed a method for the features extraction, selection, and identification of gamelan note and the proper instrument. It was an approach based on Fast Fourier Transform (FFT), and support vector machines (SVMs) for note and instrument identification. Four spectral features were selected (spectral centroid, two spectral roll-off, and fundamental frequency) as inputs for SVM. Latief et al. [6] analyzed and identified Gamelan Bonang sound spectrum. He showed a method for sound-recognition with artificial neural network back propagation concept. The artificial neural network used sigmoid activation function to all layers. Extraction was performed, as well as Fourier transform, allowing the system to do the identification of voice data Gamelan Bonang using artificial neural network back propagation concept.

Wulandari et al. [7] investigated the implementation of machine learning approach to understand statistical variations contained in gamelan signals which are relevant to onsets. The method used Elman Network which consists of one hidden layer. However, there have been lacks of studies in the past that attempt to describe the spectrum of gongs both theoretically and experimentally to better comprehend such analyses. Maclachlan [8] employed Finite Element Analysis to predict the effect of a range of variations of gong geometries on modal frequencies. The predicted frequencies of the Finite Element Analysis experiment for gong models did not match the acoustic spectra for these gongs. However, the results indicate how the near harmonic overtone spectra recorded for these gongs have been produced by the right combination of physical properties. In his concluding work [9], a series of musical gongs were designed with up to the first five overtones tuned to within 5% of the harmonic series using purposefully added prestresses generated by cold-forging small dimples. The frequencies of overtones were most sensitive to the depth of the dimples located near the maxima of modal stresses.

Ayers and Horner [10]– [11] analyzed the Gong Ageng, the largest gong used in Gamelan, using a phase vocoder spectral analysis to estimate the amplitudes and frequencies of the partials. The Gong Ageng has about a dozen prominent exponentially decaying partials, with some component frequency ratios that closely correspond to harmonics and other that are inharmonic. Many of the partials have a slow amplitude and frequency modulation of a few Hertz, and a faster modulation around 20 Hz resulting from multiple components falling into the same harmonic bin. The studies in [12]– [13] indicated that gongs show a pronounced nonlinear behavior. The pitch of the larger gong glides downward as much as three semitones after striking, whereas that of the smaller gong glides upward by about two semitones. Rossing et al. [14] investigated some vibrational modes of the larger gongs. Some of the modes were confined pretty much to the flat inner portion, some to sloping shoulders, and some involve considerable motion in both parts. When the gong was hit near the center, the central modes clearly dominate the sound. When the gong was hit lightly on the shoulder, the lowest mode is heard.

Fig. 1. The Gong Kempul

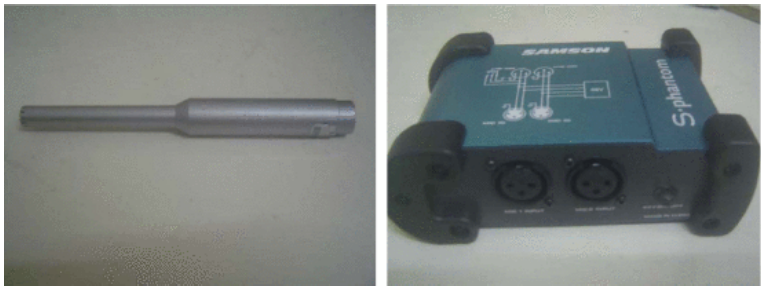

Fig. 2. Condenser microphone (left) and phantom power (right)

This paper considers the spectral and temporal properties of Javanese Gong Kempul to examine the spectral harmonics and partials of theses gongs, and how they are put together to produce the continuous beats. It was also found that the Gong Kempul possesses a degree of harmonicity, making it sounds with targeted pitch.

SECTION II.
Spectral Analysis

A. Measurement Setup

We recorded 48kHz sample tones of the Gong Kempul using impulse response measurement in ARTA. First, a measurement condenser microphone acquired the acoustic signal of the Gong Kempul, externally powered by a phantom power as shown in Figure 2. A sound card then interfaced and digitized this signal in order that computers are able to recognize.

B. Normal Modes of Membrane [15]

For any circular membrane with a fixed boundary at radius a, one possible approach to describe the free oscillations is given by:

$$\omega_{mn} = ck_{mn} = (c/a)j_{mn} \tag{1}$$

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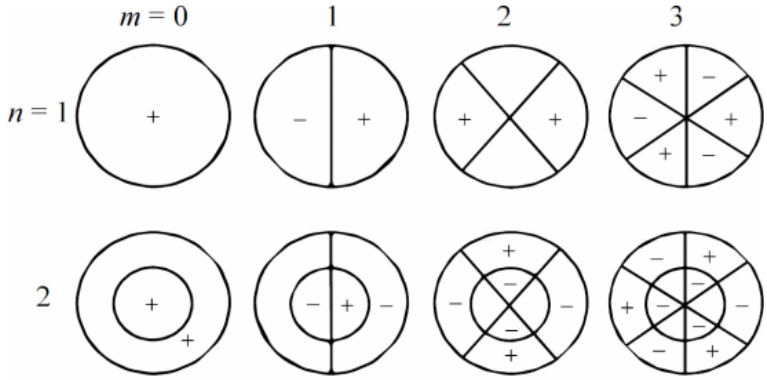
where $\omega =$ angular frequency, $c =$ speed of sound, $k =$ wave number, $j_{mn} = n^{\text{th}}$ root of Bessel function J_m These subscripts have the physical interpretation that m tells the number of nodal line going diametrically across the membrane, and n is the number of concentric nodal circles, as shown in Figure 3.



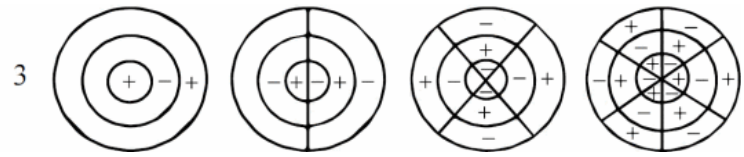


Fig. 3. Nodal lines and circles for some modes

The partials of the Gong Kempul were calculated using Eq. (1) with the total radius, including the gong shoulder, of 44 cm are shown in Table 1.

TABLE I. Calculated Partial Frequencies of the Gong Kempul

Partial at mode *m,n*	Frequency (Hz)
f_{01}	301.5 Hz
f_{11}	480.3 Hz
f_{21}	643.7 Hz
f_{02}	691.94 Hz
f_{12}	879.5 Hz
f_{22}	1055.25 Hz
f_{03}	1082.4 Hz
f_{13}	1275.3 Hz
f_{23}	1456.8 Hz

C. Spectral Properties

The plot of frequency spectrum of the Gong Kempul is shown in Figure 4 and Figure 5. Hanning window was used to smooth out the spectrum profile. The spectral peaks indicate partials with significant strength. The log frequency scale was used in both smoothed and unsmoothed frequency spectrum analyses.

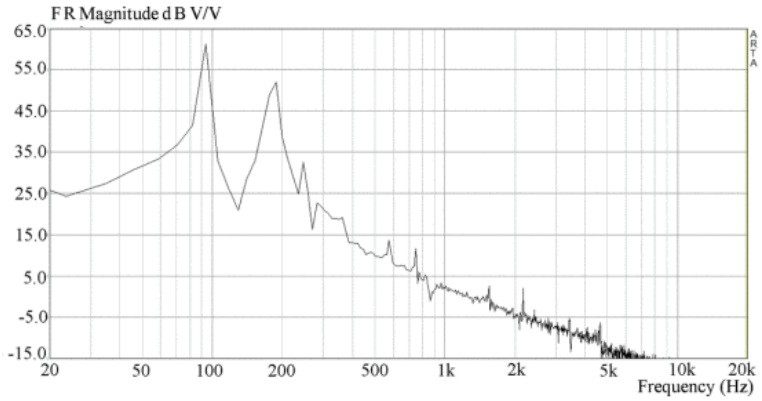


Fig. 4. Unsmoothed frequency spectrum of The Gong Kempul

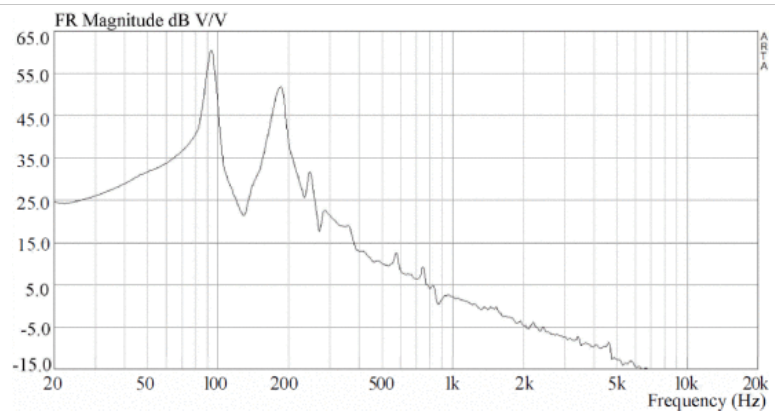


Fig. 5. Smoothed frequency spectrum of The Gong Kempul

TABLE II. Measured Partial Frequencies of the Gong Kempul

Partial #	Unsmoothed partial frequency (Hz)	Smoothed partial frequency (Hz)
1	93.8	93.8
2	187.5	187.5
3	246.1	246.2
4	281.3	281.3
5	339.8	339.8
6	375	375
7	457	457
8	574.2	574
9	750	750
10	832	831.6

Table 2 shows the ten first partial frequencies of the Gong Kempul, averaged from four trials. The partials in Table 1 do not match with that in Table 2 due to various restraining things. The measured partial frequencies in Table 2 are lower than the calculated partial frequencies in Table 1. Equation 1 perfectly holds for membranes whose tension is the only restoring force. This condition can only be satisfied in vacuum space. The decrease of frequencies might be caused by the mass of the gong and material stiffness, which in turn, give a low natural frequency as in mass-spring oscillation case. The irregularities of the Gong Kempul which stems from the manufacturing processes, including those caused by tapping the gong during tuning, also affect the partials. Furthermore, the massive and thick solid metal, in a sense, brings about inert properties of the Gong Kempul. However, the partials at 93.8 Hz, 187.5 Hz, 281.3, and 375 Hz are the strict integer multiples of the fundamental frequency. Hence, these partials are the first, second, third, and fourth harmonics, respectively. Such degree of harmonicity of the partials leads to the pitched sound of the Gong Kempul, differing it from that of Chinese gongs which sound like a crash. Therefore, the boss of the gong can bring the harmonic nature, as opposed to that calculated in Equation 1 which employs the non-strict integer multiples of roots of Bessel function. Nevertheless, some of the partials of Gong Kempul are not at strict multiples of the first harmonic. For instance, the inharmonic partial at 246.2 Hz falls between the second and third harmonics. The partials with the

most significant magnitudes are the first and second harmonics, whereas the others do not seem to be significant. The harmonic and or inharmonic partials that occur in fairly close frequency beat together and form the roaring sound which is often associated with Bima's laughter (Bima is one of the Pandawa brothers in the story of Mahabarata).

Figure 6 show the burst decay graph to investigate the most prominent partials along a certain time intervals at the opening and tail of the sound, respectively. A complex Morlet wavelet analytic signal is used in convolution with system impulse response. The magnitude of that response, also known as *wavelet scalogram*, represents the envelope of the shaped burst response decay. A complex Morlet wavelet analytic signal is defined as:

$$w(t) = e^{-t^2/\tau^2}(\cos\omega_o t + j\sin\omega_o t) \tag{2}$$

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The Fourier transform of the Morlet wavelet is equal to:

$$W(\omega) = e - (\omega - \omega_o)^2\tau^2/4 \tag{3}$$

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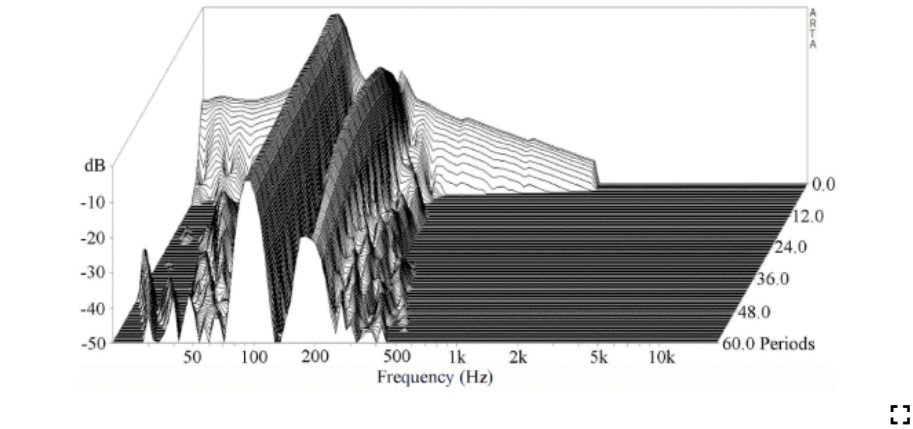

Fig. 6. The burst decay graph at the beginning of the sound

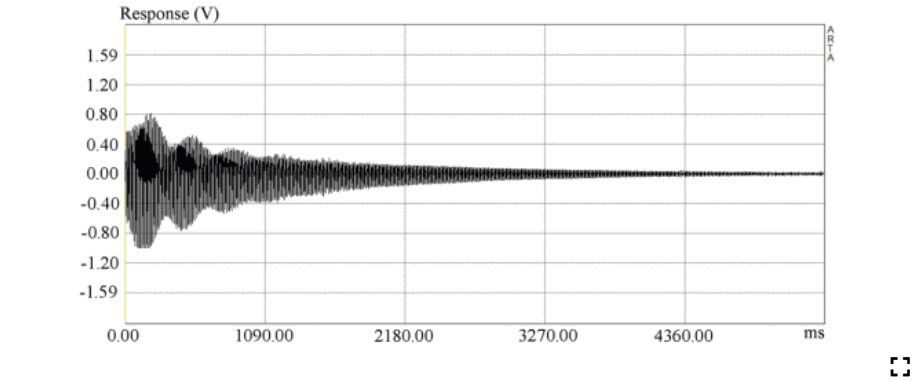

Fig. 7. Impulse Response

It can be seen that the fundamental frequency at 93.8 Hz decays much more slowly than the other harmonics and increases in dominance as the other partials weaken in decaying tone.

SECTION III.

Impulse-generated Response

A short impulse can be represented as the instantaneous and gentle strike on the Gong Kempul boss. Thus, an impulse-generated response was recorded to analyze in the time domain. Figure 7 shows the impulse response of the Gong Kempul within about 5400 ms in which the sound vanishes. The time plot specifies three-cycle beat frequency indicating three waves (Javanese: ombak telu).

Upon zooming the time plot, as shown in Figure 8(a)-(c), we can observe that the Gong Kempul starts withs a repetitive wave of 11.33 ms period. After a while, the repetitive period decreases to 11.00 ms. Therefore, the gong starts with frequency of 88.26 Hz and subsequently the frequency shifts to 90.90 Hz. There is a difference of 2.64 Hz which is the origin of "ombak telu". At the tail of wave repetition, the period is 10.67 ms which coincides with the fundamental frequency, 93.8 Hz. In the other words, the sound fundamental wavelength can be derived by the following formula,

$$\lambda = \frac{343 + (T - 20) \times 0.6}{f} \tag{4}$$

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where T is the ambient temperature during measurement (°C), f= =frequency(Hz), yielding the sound fundamental wavelength is 3.67 m with $T = 26$ degree Celsius.

Figure 9 shows the impulse response envelope, also called Energy-Time Curve. The envelope of signal x(t) is an envelope of signal absolute values. Mathematically, it is defined by the following expression:

$$e(t) = \left[x^2(t) + \bar{x}^2(t)\right]^{1/2} \tag{5}$$

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where is $\bar{x}(t)$ is the Hilbert transformation of $x(t)$, given by:

$$\bar{x}(t) = \frac{1}{\pi} x(t) \otimes \frac{1}{t} \tag{6}$$

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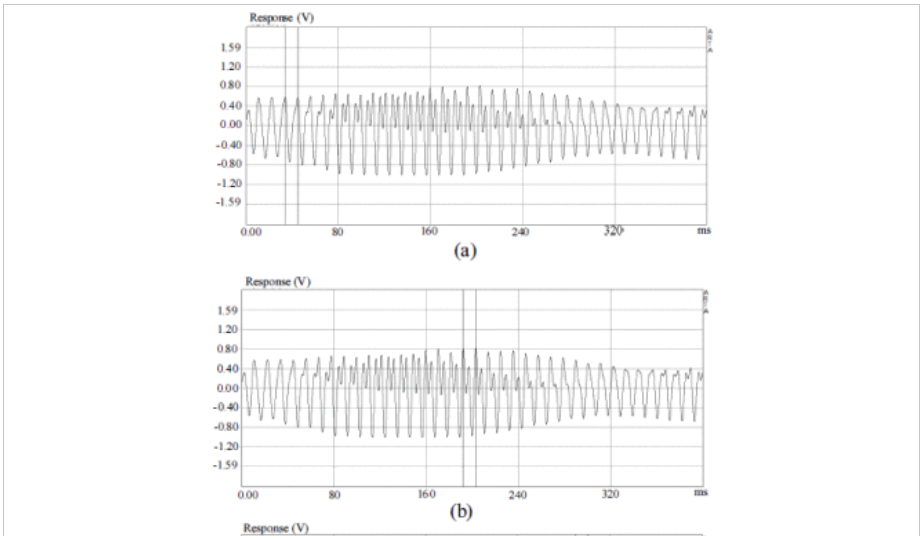
The overall shape is a nearly-linear decay where the energy peak out at 151.938 ms. The beat phenomenon are apparent for about 1233 ms and such modulations are clearly found for the first five harmonics.



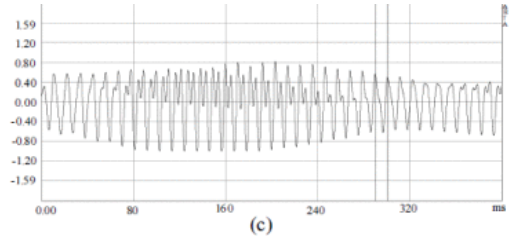


Fig. 8. Repetition waves with varying periods at the (a) beginning (b) middle and (c) end of the wave

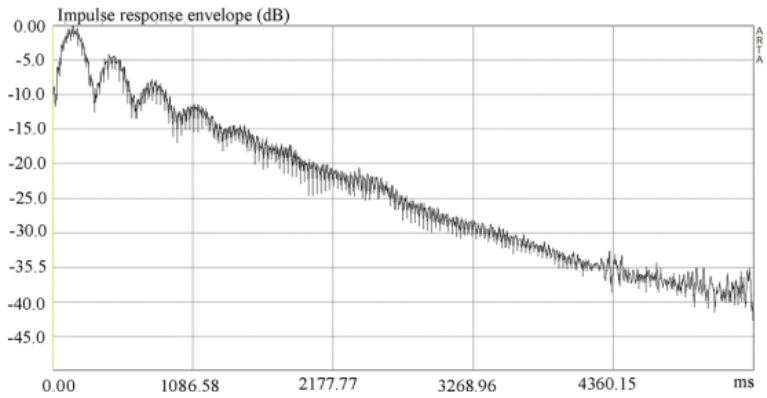


Figure 9 Impulse response envelope

SECTION IV.
Conclusion

The 56 cm Gong Kempul has a nature of harmonicity with fundamental frequency at 93.8 Hz, while the second, third, and fourth harmonics are, respectively, at 187.5 Hz, 281.3, and 375 Hz with several partials lie between each harmonics. It can also be inferred that the fundamental frequency at 93.8 Hz decays much more slowly than the other harmonics and increases in dominance as the other partials weaken. The wave-like sound of the Gong Kempul is due to the beat phenomenon between two partials that occur closely. The tuning of the Gong Kempul was carried by tapping the outer part of the gong which affects the vibration modes. In the future, gongs in different size and materials are about to be investigated.

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